of incidence of light. **According to laws of reflection,**

(i) The incident ray, reflected ray and normal to the reflecting surface are all coplanar (ie. lie in the same plane).

(ii) The angle of incidence *i* is equal to the angle of reflection *r*.

$$i = r \qquad (6.1)$$

The law of reflection is shown in Figure 6.1.

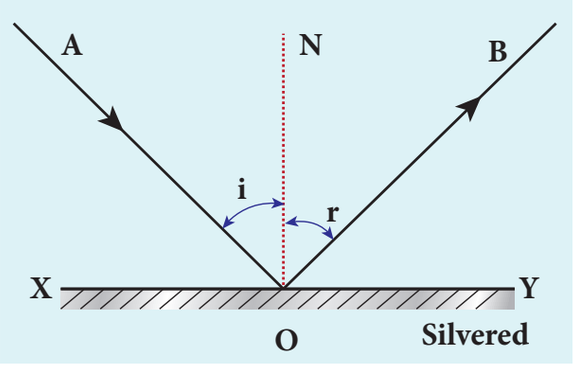
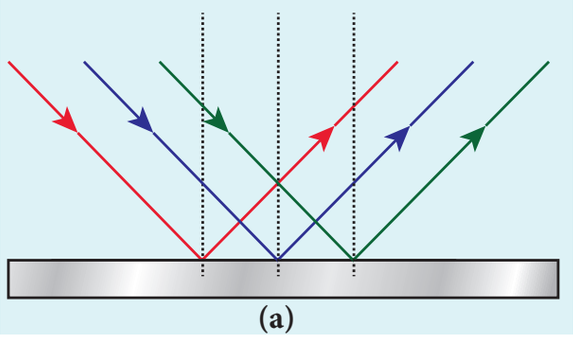


Figure 6.1 Reflection of light

The laws of reflection are valid at each point for any reflecting surface whether the surface is flat (or) curved. If the reflecting surface is flat, then incident parallel rays after reflection come out as parallel rays as shown in Figure 6.2(a). If the reflecting surface is irregular, then the incident parallel rays after reflection come out as irregular rays (not parallel rays). Still the laws of reflection are valid at every point of incidence in irregular reflection as shown in Figure 6.2(b).



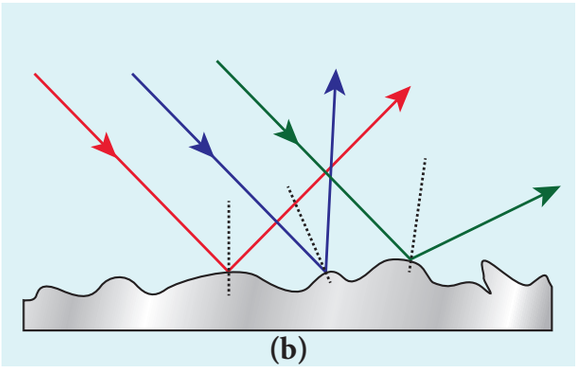


Figure 6.2 (a) Regular and (b) irregular reflections

6.1.3 Angle of deviation due to reflection

The angle between the direction of incident ray and the reflected ray is called *angle of deviation due to reflection*. It is calculated by a simple geometry as shown in Figure 6.3(a). The incident light is AO. The reflected light is OB. The un-deviated light is OC which is the continuation of the incident light. The angle between OB and OC is the angle of deviation d. From the geometry, it is written as, $d = 180 - (i+r)$. As, $i = r$ in reflection, we can write angle of deviation in reflection as,

$$d = 180 - 2i \qquad (6.2)$$

The angle of deviation can also be measured in terms of the glancing angle α which is measured between the incident ray AO and the reflecting plane surface XY as shown in Figure 6.3(b). By geometry, the angles $\angle AOX = \alpha$, $\angle BOY = \alpha$ and $\angle YOC = \alpha$ (all are same). The angle of deviation d is the angle $\angle BOC$. Therefore,

$$d = 2\alpha \qquad (6.3)$$

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